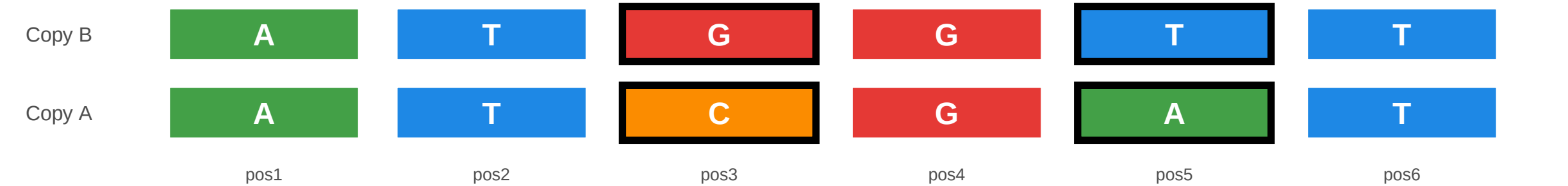


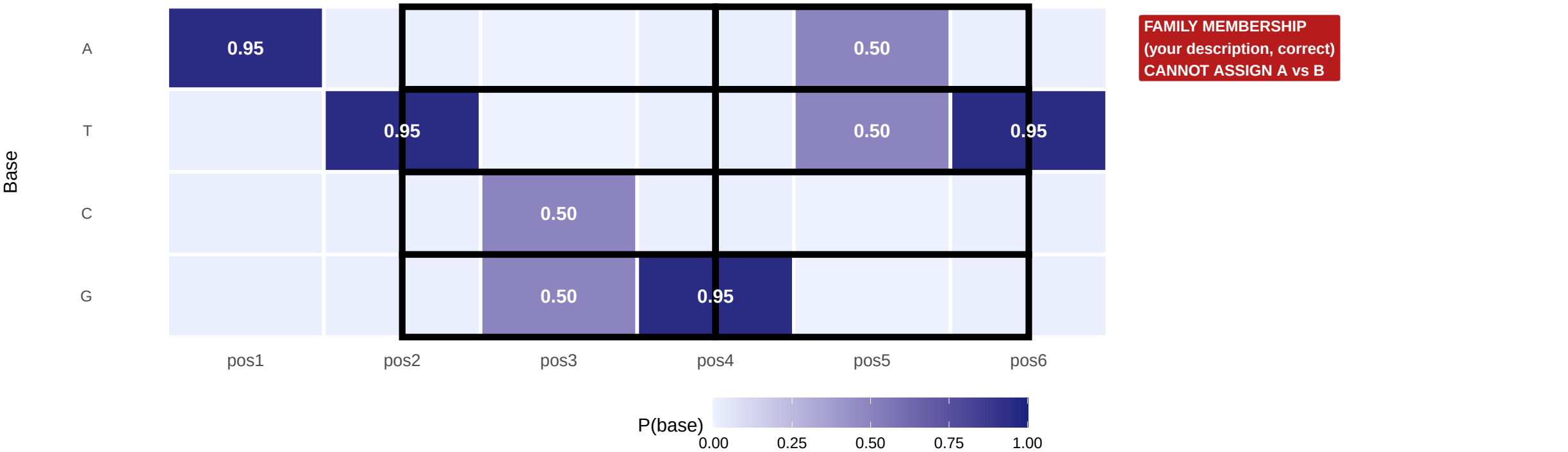
Classical (family) profile HMM vs. per-copy profile HMMs

Addressing your first concern. Toy 2-copy, 6-base example. No mention of any other concept.

Toy example: two copies of length 6, SNPs at pos 3 and pos 5

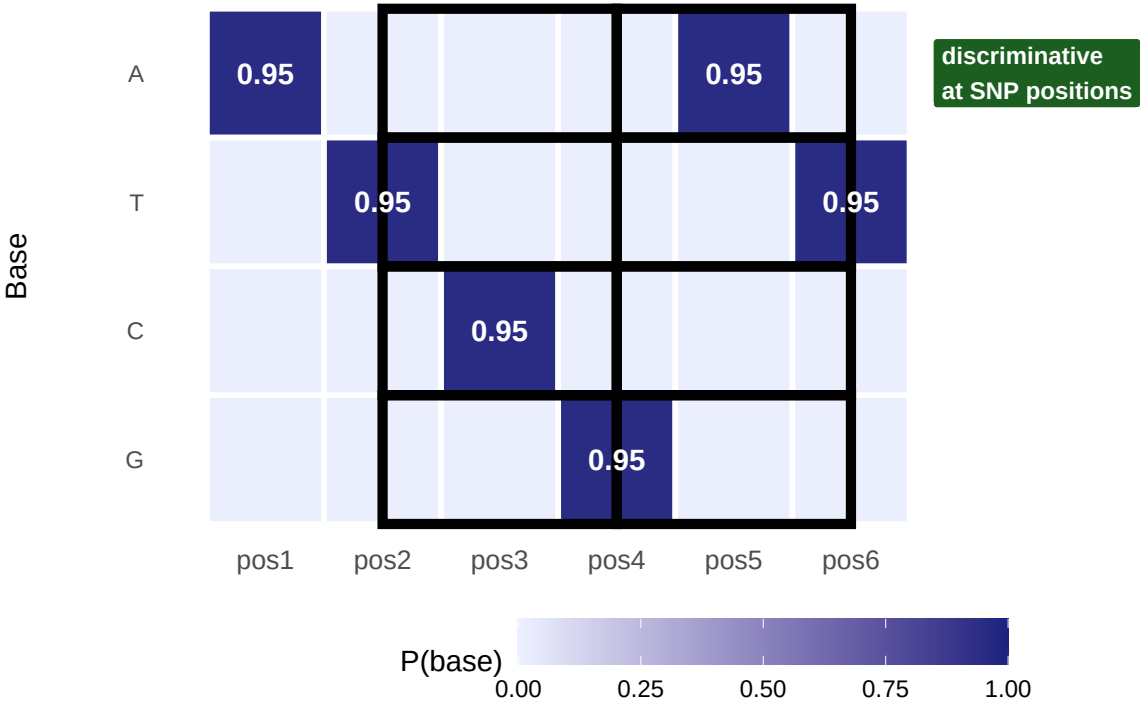


1. Classical profile HMM (built from MSA of A and B)

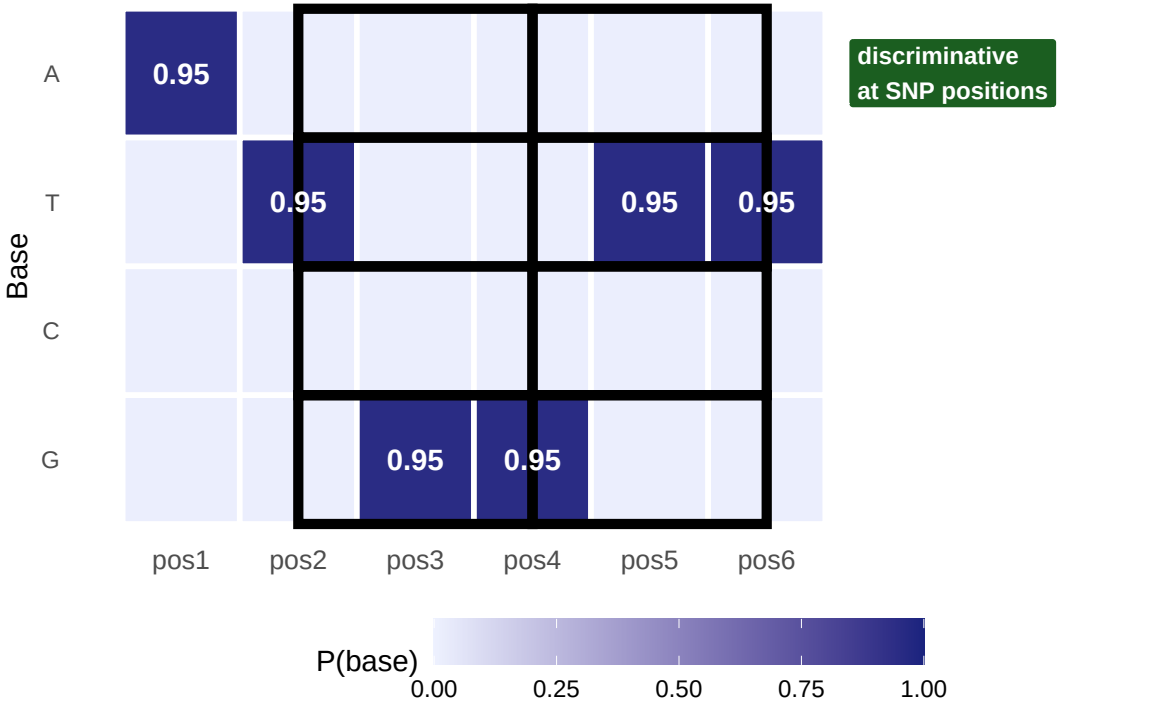


Built from the alignment (MSA) of copies A and B (this is what you described in your email).
At SNP positions, the match state has 0.50 / 0.50 emission for the two alleles (averaging across the MSA).
Viterbi gives P(read fits family). There is only ONE model -> no way to compare P(read | A) vs P(read | B).
-> Useful for asking 'does this sequence belong to the family?' NOT useful for assigning a read to a specific copy.

2a. Per-copy profile HMM for Copy A



2b. Per-copy profile HMM for Copy B



Build ONE profile HMM per copy, from that copy's sequence alone.
At every position, match state emits 0.95 for that copy's base, 0.017 for any other base.
Now there are TWO models. For a read R: $\log L_A = \log P(R | \text{HMM}_A)$, $\log L_B = \log P(R | \text{HMM}_B)$.
 $D\log L = \log L_A - \log L_B$ -> sign tells us which copy the read came from.
-> Solves the read-to-paralog assignment problem.